

The Normal Curve

The Normal Distribution is bell-shaped, and we will draw a density curve on the graph to the right. You can see that this curve will approximate the data, not describe it perfectly.

Density curves, like distributions, come in many shapes. A density curve is often a good description of the overall _____ of a distribution.

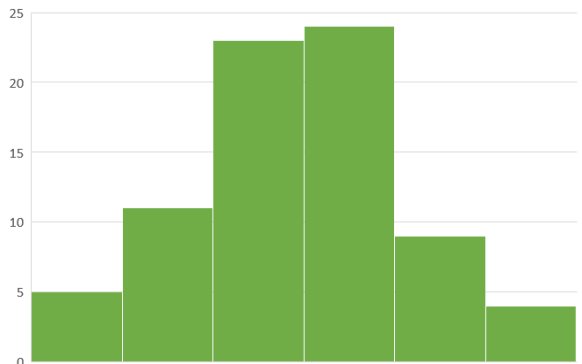
A density curve is an approximation that makes it easy for use.

What data is “normal”?

Normal distributions (where we fit the normal model to a set of data) are appropriate for many distributions whose shapes are unimodal and approximately symmetric.

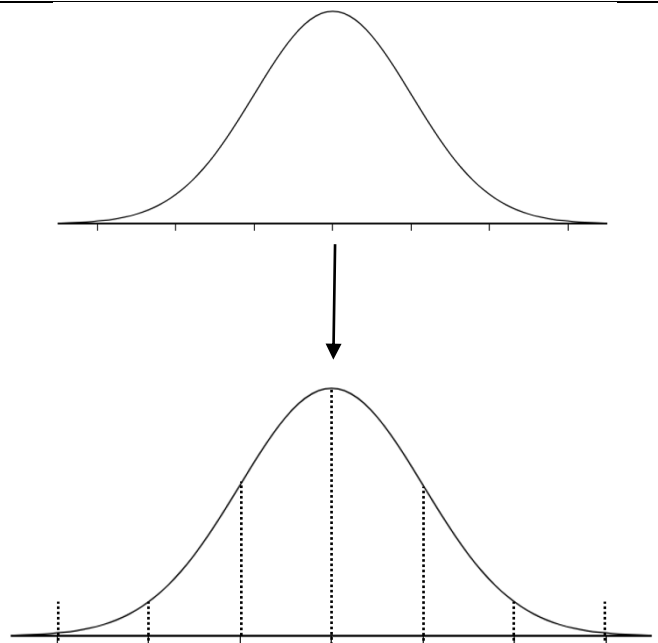
The Normal Distribution

Mean = _____ Standard Deviation = _____ which is described as _____



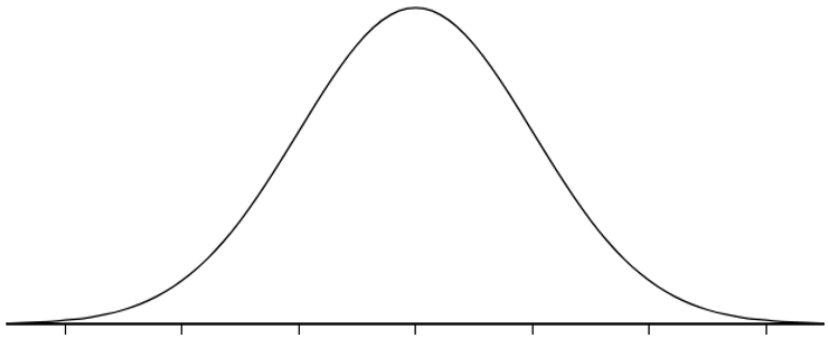
A Normal Distribution density curve has the following properties:

- Symmetric around the _____
- _____ = _____ = _____
- 50% of observations are less than the _____
- 50% of observation are greater than the _____
- The standard deviation tells us how measurements for a group of observations are spread out from the _____
- In a normal distribution, approximately all of the data lies within _____ standard deviations above and below the mean.



Example

The heights of American women aged 18 to 24 years old with a mean of 65.6 inches and a standard deviation of 2.5 inches. Assuming the population is normally distributed, draw a normal curve and label 3 standard deviations out on either side.

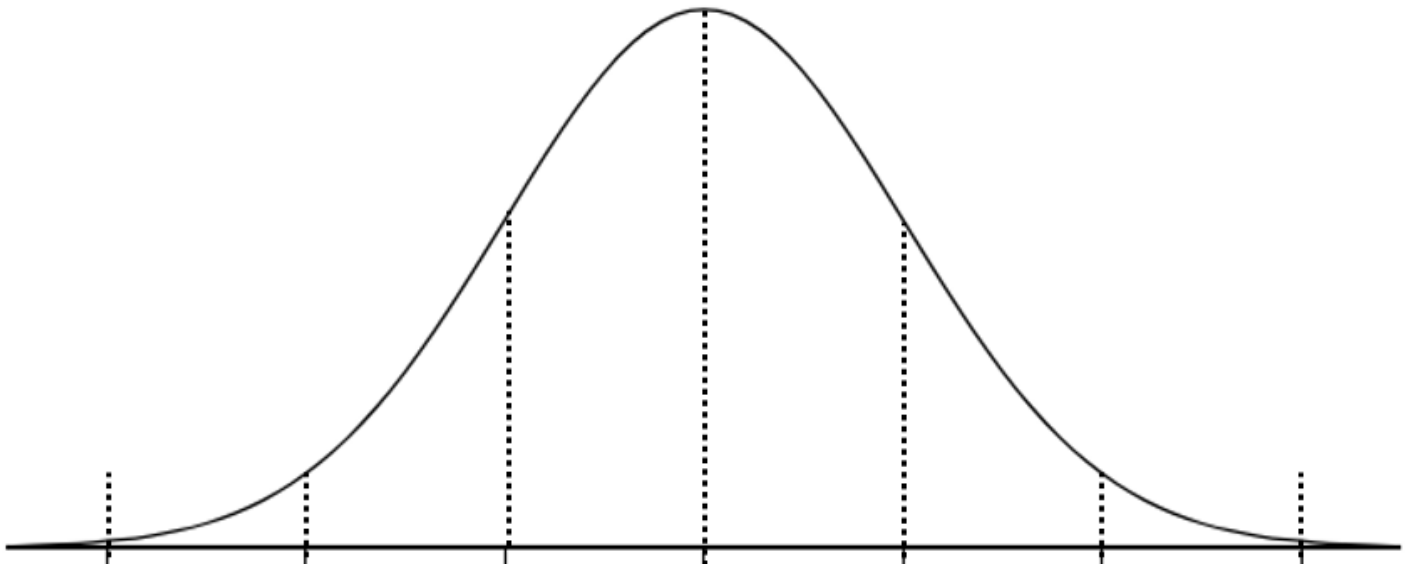


The Empirical Rule

In a normal distribution, we can describe the variability (_____) of the data in terms of probabilities:

- _____ of data is likely to be within _____ standard deviation around the mean
- _____ of data is likely to be within _____ standard deviation around the mean
- _____ of data is likely to be within _____ standard deviation around the mean

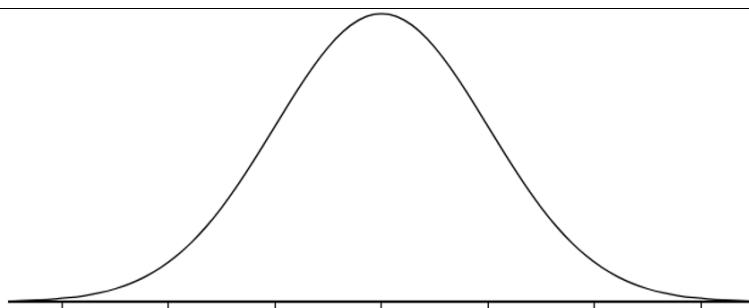
These three percentages _____ describe the Empirical Rule. These approximations can be used with the symmetry of a normal distribution to approximate proportions of data.



Directions: Draw a normal distribution and use the Empirical Rule to answer each question.

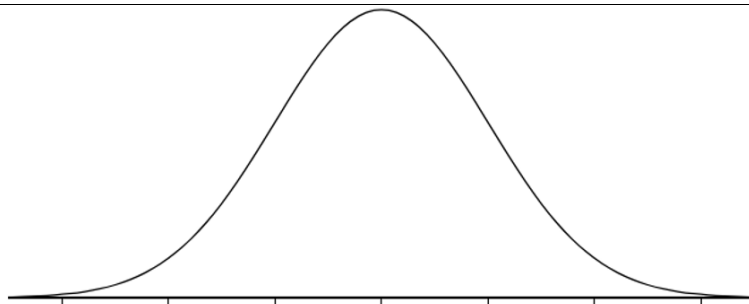
The mean test score on the Unit 1 Assessment was an 82 with a standard deviation of 4.

95% of the test scores were between what two scores?

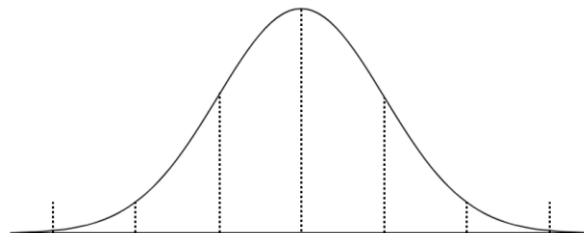
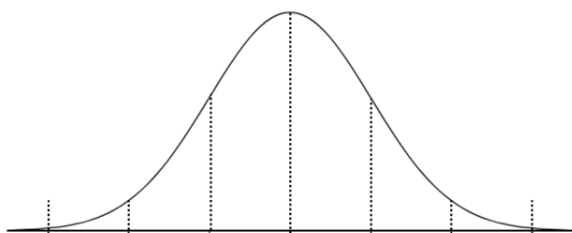


The mean age at the concert was 38 years old with a standard deviation of 5.5 years.

What percent of the concert goers were between 32.5 years old and 43.5 years old?



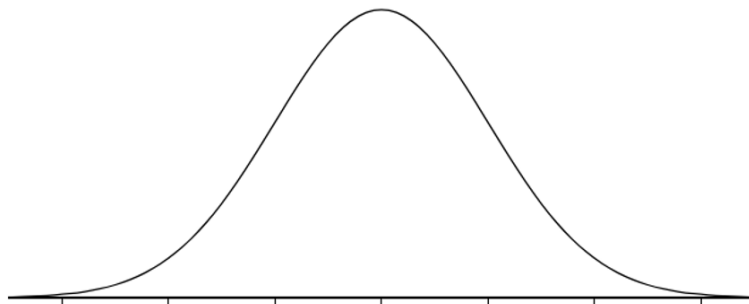
We can break down the Empirical Rule further to calculate more proportions of the distribution.



The mean test score on the Unit 1 Assessment was a 82 with a standard deviation of 4.

1) What percent of test scores fell below a 74?

2) What percent of test scores fell above a 78?



Standardizing

Mallory took a physics test and got an 86%. How did she do?

If the class mean was 97%, did she do well? Why? _____

If the class mean was 62%, did she do well? Why? _____

The location of an observation in relationship with its mean is one that is asked often. So often, in fact, we have a method of calculating how far away an observation is from its mean. This method is called *standardizing*, and it is the process of counting how many standard deviations above or below the mean an observation falls.

On this physics test, Mallory scored an 86% and on the next test, Mallory scored an 82%. Which was a more impressive score? We would want to know how Mallory did relative to the class!

We use standardizing to compare two things that have a different scale.

- On her first test, Mallory scored an 86% and the class mean was 83% and the standard deviation was 4.5%
- On her second test, Mallory scored an 82% and the class mean was 78% and the standard deviation was 5.2%.

Subtracting her scores from each of the means tells us the distance from the mean, but to determine if this distance is “typical”, we divide by the standard deviation to see how many standard deviations it is away from the mean. The resulting number – how many standard deviations away from the mean an observation is – is called a **z-score**.

First Test

Second Test

Interpreting:

- A z-score of _____ means that Mallory _____
- A z-score of _____ means that Mallory _____

Z-Scores

If x is an observation from a distribution that has known mean and standard deviation, the z-score for x is:

	• A z-score tells us how many standard deviations from the _____ an observation falls and in what direction. • Observations larger than the mean have _____ z-scores. • Observations smaller than the mean have _____ z-scores.
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Examples

1) Female shoe sizes are approximately normally distributed with a mean of 8 and a standard deviation of 0.75. Find and interpret the z-score of a female with a size 6 shoe.

2) Joe is 74" tall. The average height of an adult male is 70" with a standard deviation of 2.5". Find and interpret the z-score for Joe's height.

3) Corbin scored 1100 on the SAT in 2020. The distribution of SAT scores in 2020 was normally distributed, with a mean score of 1083 and a standard deviation of 194. Corbin also took the ACT in 2020 and earned a score of 23. The 2020 ACT scores were normally distributed with a mean of 21 and a standard deviation of 1.62. On what standardized test did Corbin score better on?